

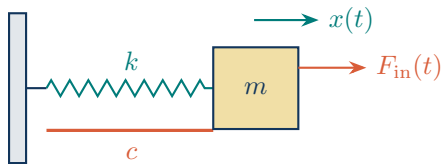
# Spring–Mass–Damper Dynamics

Equations, variables, and power balance used in the simulation

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Based on `spring_mass_damper.py`

# Physical model and sign convention



## Convention used by the script

Positive displacement and applied force point to the right. Positive power is power delivered *to the mass*.

The spring and damper forces exerted on the mass oppose displacement and velocity:

$$F_s = -kx, \quad F_d = -c\dot{x}.$$

# Equation of motion

## Forced linear model

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F_{\text{in}}(t)$$

Newton's second law is evaluated in the code as

$$\ddot{x} = a = \frac{F_{\text{in}} - c\dot{x} - kx}{m}.$$

## Force balance

$$F_{\text{net}} = F_{\text{in}} + F_s + F_d = ma.$$

## Free response

When  $F_{\text{in}} = 0$ :

$$m\ddot{x} + c\dot{x} + kx = 0.$$

## Variables and default values

| Symbol                         | Meaning                              | Script default (SI)                  |
|--------------------------------|--------------------------------------|--------------------------------------|
| $m$                            | Mass                                 | 1.0 kg                               |
| $c$                            | Viscous damping coefficient          | 0.5 N s/m                            |
| $k$                            | Spring stiffness                     | 10.0 N/m                             |
| $x, v = \dot{x}, a = \ddot{x}$ | Displacement, velocity, acceleration | $x(0) = 0, v(0) = 0$                 |
| $t$                            | Simulation time                      | $0 \leq t \leq 20$ s                 |
| $F_{\text{in}}$                | Externally applied force             | pulse amplitude $kx_{\text{target}}$ |
| $x_{\text{target}}$            | Pulse target displacement            | 0.05 m                               |
| $t_p$                          | Pulse duration                       | 1.0 s                                |

### Input used in the script

$$F_{\text{in}}(t) = \begin{cases} kx_{\text{target}}, & t \leq t_p, \\ 0, & t > t_p. \end{cases} \Rightarrow F_{\text{in}} = 0.5 \text{ N during the pulse.}$$

# First-order state representation

The ODE solver advances the state vector

$$\mathbf{z} = \begin{bmatrix} x \\ v \end{bmatrix}, \quad \dot{\mathbf{z}} = \begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ \frac{F_{\text{in}} - cv - kx}{m} \end{bmatrix}.$$

Equivalently,

$$\dot{\mathbf{z}} = \underbrace{\begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}}_A \mathbf{z} + \underbrace{\begin{bmatrix} 0 \\ 1/m \end{bmatrix}}_B F_{\text{in}}.$$

## Numerics

The script uses `scipy.integrate.solve_ivp` with RK45 at 100001 sampled time points,  $rtol = 10^{-9}$ , and  $atol = 10^{-11}$ .

# Instantaneous power

For a force acting on the mass,  $P = Fv$ . The implementation therefore calculates:

$$P_{\text{spring}} = F_s v = -kxv,$$

$$P_{\text{damper}} = F_d v = -cv^2 \leq 0,$$

$$P_{\text{input}} = F_{\text{in}} v,$$

$$P_{\text{net}} = F_{\text{net}} v = mav = \frac{dK}{dt}.$$

## Damper loss rate

$$P_{\text{dissipated}} = -P_{\text{damper}} = cv^2 \geq 0.$$

## Code convention

A negative component power means that component removes energy from the mass.

# Stored energy and balance

## Energy definitions

$$K = \frac{1}{2}mv^2, \quad U = \frac{1}{2}kx^2,$$

$$E_{\text{mech}} = K + U.$$

Their rates are

$$\frac{dK}{dt} = mav, \quad \frac{dU}{dt} = kxv.$$

## Power balance

$$\frac{dE_{\text{mech}}}{dt} = \frac{dK}{dt} + \frac{dU}{dt} = P_{\text{input}} + P_{\text{damper}}.$$

Because  $P_{\text{damper}} \leq 0$ , damping drains mechanical energy.

## Integrated balance checked by the script

$$E_{\text{mech}}(t) + E_{\text{diss}}(t) = E_{\text{mech}}(0) + E_{\text{input}}(t),$$

where  $E_{\text{diss}} = \int_0^t cv^2 d\tau$  and  $E_{\text{input}} = \int_0^t F_{\text{in}} v d\tau$ .

# Active and reactive mechanical power

## Mechanical analogy used in the script

$$P_{\text{active}} = P_{\text{dissipated}} = cv^2$$

$$P_{\text{reactive}} = \frac{dU}{dt} + \frac{dK}{dt}$$

- The damper is resistive: it irreversibly converts mechanical energy to heat.
- The spring and mass are storage elements: their energy changes reversibly between  $U$  and  $K$ .
- The script plots both quantities alongside instantaneous input power and stored/dissipated energy.



## Key response properties

$$\omega_n = \sqrt{\frac{k}{m}}, \quad c_{\text{crit}} = 2\sqrt{km}, \quad \zeta = \frac{c}{c_{\text{crit}}}.$$

For  $\zeta < 1$ :

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}.$$

### Defaults evaluated

$$\begin{aligned}\omega_n &= 3.1623 \text{ rad/s}, \\ c_{\text{crit}} &= 6.3246 \text{ N s/m}, \\ \zeta &= 0.0791.\end{aligned}$$

Underdamped response

The pulse injects energy; the spring and mass exchange it; the damper dissipates it.

## Explore the interactive simulation

Adjust the model parameters and inspect the response, forces, power, and energy in the live Streamlit app.

[springmassdamperurp.streamlit.app](https://springmassdamperurp.streamlit.app)

Scan the QR code or follow the link to open the simulation.

